DP-900
AZURE DATA

COURSE SLIDES - WSQ

Microsoft Azure Data Fundamentals (DP-900)

WSQ Course Code: TGS-2023036641

Trainer: Dr. Alfred Ang

Conducted by Tertiary Infotech Academy Pte Ltd - UEN 201200696W

Version v1.0 - 8 July 2026

Course Administration

Welcome & Housekeeping

Digital Attendance (Mandatory)

1. Take AM, PM and Assessment digital attendance for WSQ-funded courses.
2. The trainer displays the SSG TRAQOM digital attendance QR code.
3. Scan the QR code with your mobile phone and submit your attendance.
4. Minimum 75% attendance is required for assessment and funding.

About the Trainer

?

Your Trainer

General Trainer template - to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer, Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

CERTIFICATION
Microsoft Azure certified cloud and data trainer.

DELIVERS
WSQ courses on Azure, AI, data fundamentals and automation.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Icebreaker

1. Your name and organisation or role.
2. Your experience with data, databases, reporting or Azure.
3. One DP-900 topic you want to understand better by the end of class.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively - no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

SCHEDULE

Lesson Plan - 2 Days

Day 1

Admin briefing

Core data concepts

Relational concepts

Azure SQL and open-source databases

Day 2

Azure Storage and Cosmos DB

Analytics services

Power BI

Capstone review and assessment

Timing

9:00 AM-6:00 PM

Lunch and tea breaks

Final assessment from 4:00 PM

WHAT YOU WILL ACHIEVE

Learning Outcomes

1
LO1
Describe structured, semi-structured and unstructured data.

2
LO2
Compare transactional and analytical workloads.

3
LO3
Identify common data roles and data store options.

4
LO4
Explain relational concepts, SQL, normalization and database objects.

5
LO5
Describe Azure SQL and open-source database services on Azure.

6
LO6
Describe Azure Storage, Azure Cosmos DB and NoSQL patterns.

7
LO7
Explain ingestion, processing, analytical data stores, batch and streaming analytics.

8
LO8
Describe Microsoft Fabric, Azure Databricks, real-time analytics and Power BI concepts.

ASSESSMENT RULES

Assessment Briefing

1. Clear your table and keep only approved materials.
2. No photos, recording or discussion during assessment.
3. Open book: use the slides, Learner Guide and approved notes.
4. Complete Written Assessment and Practical Performance as instructed.
5. Submit answers on the LMS when time is up.

FINAL ASSESSMENT

Assessment and Funding

1. Written Assessment (WA): 1 hour, open book.
2. Practical Performance (PP): 1 hour, open book.
3. Assessment starts on Day 2 at 4:00 PM.
4. Minimum 75% attendance is required for funding eligibility.
5. Courseware and assessment: https://lms-tms.tertiaryinfotech.com/

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM: scan the QR code on the LMS
2. Assessment digital attendance
3. Sit WA then PP, open book
4. Submit answers on the LMS
5. Sign the Assessment Summary Record

LAB ACCESS

Access the Hands-On Labs

1. GitHub repository: https://github.com/tertiarycourses/TGS-2023036641---Microsoft-Azure-Data-Fundamentals-DP-900-Course-Code-TGS-2023036641 Open DP-900 lab repository
2. Option A: git clone the repository and open the labs folder.
3. Option B: use GitHub Code > Download ZIP.
4. Complete labs in order because later labs reuse earlier concepts.

DP

COURSE CONTENT

Microsoft Azure Data Fundamentals

COURSE MAP

DP-900 Exam Skill Areas

1 **Core Data Concepts**
Data types, workloads, roles, stores and file formats

2 **Relational Data on Azure**
Relational design, SQL, Azure SQL and open-source databases

3 **Non-Relational Data on Azure**
Azure Storage, Blob, Files, Tables and Cosmos DB

4 **Analytics Workloads on Azure**
Ingestion, processing, analytical stores, Fabric, Databricks and real-time analytics

5 **Power BI and Exam Readiness**
Models, visualizations, service mapping and capstone review

LABS 01-10 | CONCEPT MAP

DP-900 Data Platform Flow

Transactional apps
Lab 01

>

Operational stores
Azure SQL | Cosmos DB
Labs 03-06

>

Storage layer
Blob | Files | Tables
Lab 05

>

Ingestion + processing
Pipelines | batch | stream
Labs 07-08

>

Analytical store
Lakehouse | warehouse
Lab 08

>

Power BI reports
Lab 09

This diagram follows the labs: classify data and workloads, choose operational and non-relational stores, ingest and process analytics data, then model and visualize it in Power BI.

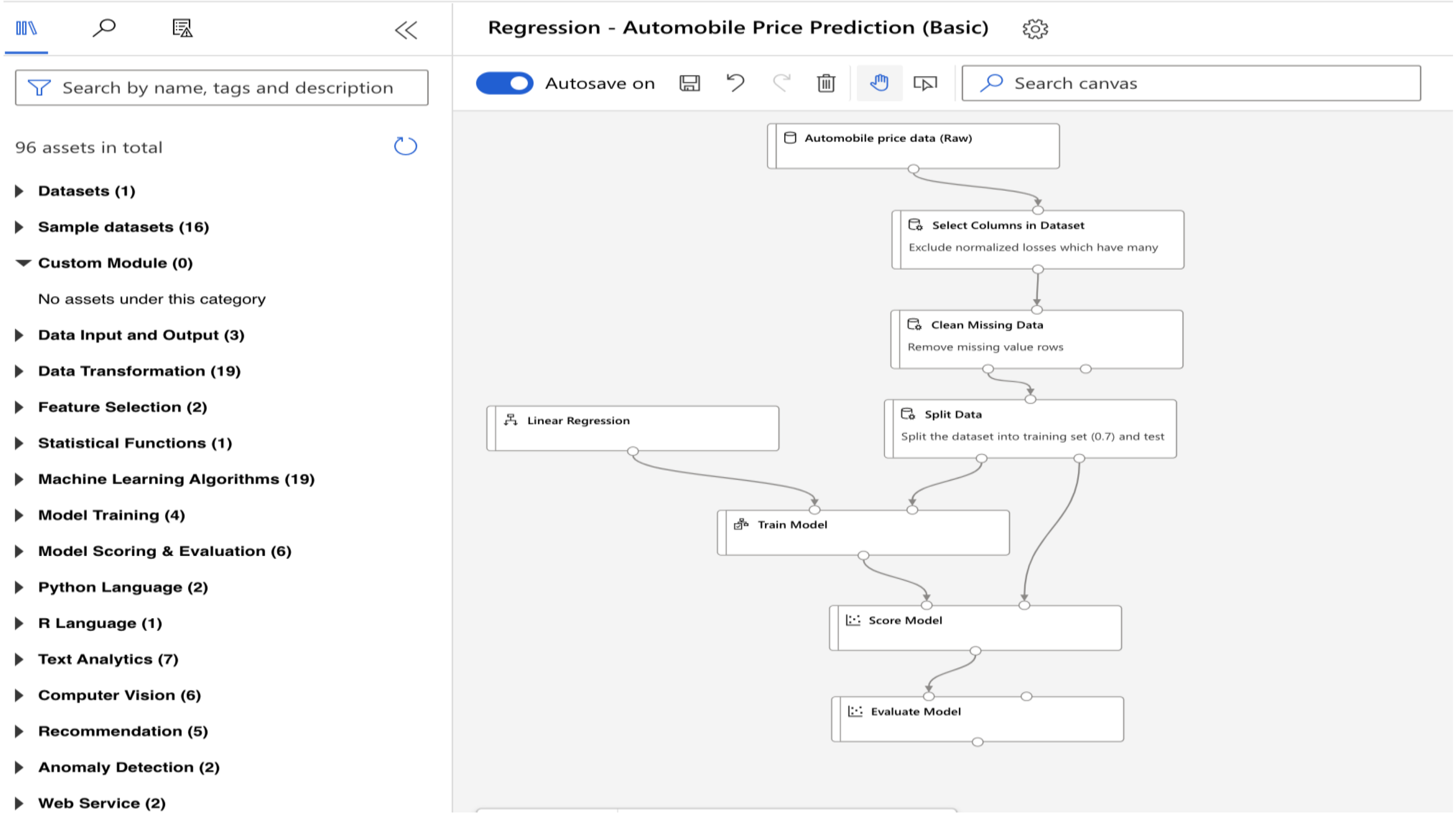
SERVICE RECOGNITION

Lab-to-Service Map

Labs 01-02	**Core concepts, roles, file formats**	>	Choose the right workload and storage pattern
Labs 03-04	**Relational and Azure SQL services**	>	Recognize relational service options
Labs 05-06	**Azure Storage and Cosmos DB**	>	Match non-relational data to services
Labs 07-08	**Analytics platforms**	>	Explain ingestion, processing and analytical stores
Labs 09-10	**Power BI and capstone**	>	Map scenarios to services and prepare for DP-900

IMPORTED DIAGRAM

Reference Visual from Existing Master PPT



Why this matters

The AI-900 master deck includes Azure pipeline visuals. DP-900 learners use the same mental model when reading ingestion, processing, model/reporting and evaluation paths.

Use it as a visual bridge into analytics services, not as a deep machine learning lesson.

01

TOPIC 01

Core Data Concepts

Data types, workloads, roles, stores and file formats

Key Concepts - Core Data Concepts

1 **Data representation**

Structured, semi-structured and unstructured data require different storage and processing patterns.

2 **Workload intent**

Transactional workloads support operations; analytical workloads support insight and reporting.

3 **Data roles**

DBAs, data engineers and data analysts own different parts of the data lifecycle.

4 **Storage choices**

File, relational, key-value, document and analytical stores solve different business problems.

WHAT YOU WILL DO

Hands-On Labs - Core Data Concepts

Lab 1

Core Data Concepts, Data Types, Workloads
Distinguish structured, semi-structured, and unstructured data.
You should have a data type table, workload table, classification answers, and g

Lab 2

Data Roles, Storage Options, File Formats
Identify data workload roles and responsibilities.
You should have role notes, file format comparison, storage mapping, and a platf

TOPIC 01 | HANDS-ON

Lab 1: Core Data Concepts, Data Types, Workloads

LAB 01

Core Data Concepts

A retail company stores sales transactions, customer profiles, product images, web logs, invoices, and analytics reports. You must classify the data and decide which workloads are transactional or analytical.

Objectives

Distinguish structured, semi-structured, and unstructured data. | Compare transactional and analytical workloads. | Identify common data store use cases.

Validation

You should have a data type table, workload table, classification answers, and glossary.

GUIDED LAB FLOW

Lab 1: Step-by-Step

1

Classify Data Types

Create a table: | Data Example | Type | Reason |

| --- | --- | --- |

| Sales order table | Structured | Fixed rows and columns |

| JSON web event | Semi-structured | Flex

2

Compare Workloads

Create a table: | Workload | Purpose | Example |

| --- | --- | --- |

| Transactional | Supports day-to-day operations | Online order processing |

| Analytical | Supports

3

Identify Workload Characteristics

For each scenario, choose transactional or analytical: 1. Insert a customer order.

2. Update inventory quantity.

3. Analyze five years of sales data.

4. Create a Power BI

4

Create a Glossary

Define: Database

Table

File

Data lake

Data warehouse

Transactional workload

Analytical workload

CHECKPOINT

Lab 1: Validate and Review

Test it

You should have a data type table, workload table, classification answers, and glossary.

Checkpoint questions

1. What is structured data?
2. What is semi-structured data?
3. Why is unstructured data common in modern systems?
4. How does an analytical workload differ from a transactional workload?

Exam focus: DP-900 frequently tests the difference between data representations and workload types.

Lab 2: Data Roles, Storage Options, File Formats

LAB 02

Core Data Concepts

The retail company is forming a data team. Management asks which people and services are needed for databases, pipelines, analytics, and reports.

Objectives

Identify data workload roles and responsibilities. | Compare common file formats. | Match storage options to use cases.

Validation

You should have role notes, file format comparison, storage mapping, and a platform diagram.

GUIDED LAB FLOW

Lab 2: Step-by-Step

1

Identify Data Roles

Create a table: | Role | Main Responsibility |
| --- | --- |
| Database administrator | Manage database availability, performance, backup, and security |
| Data engineer

2

Compare File Formats

Format	Common Use
CSV	Simple tabular exchange
JSON	Semi-structured API and event data
Parquet	Columnar analytics data
XML	Struct

3

Match Storage Options

Requirement	Possible Store
Store images and backups	Blob storage
Shared file access	Azure Files
Key-value style entities	Azure Table S

4

Draw a Simple Data Platform

Use diagrams.net: Source systems -> Ingestion -> Storage -> Processing -> Analytics -> Visualization

5

Record Service Selection Rules

Write one sentence for when to use relational, non-relational, object storage, and analytical stores.

CHECKPOINT

Lab 2: Validate and Review

Test it

You should have role notes, file format comparison, storage mapping, and a platform diagram.

Checkpoint questions

1. What does a data engineer do?
2. Why is Parquet common in analytics?
3. When would Blob storage be appropriate?
4. Why do analytics platforms separate storage and compute?

Exam focus: Know the roles and responsibilities for data workloads and common storage formats.

TOPIC RECAP

Recap - Core Data Concepts

1. DP-900 frequently tests the difference between data representations and workload types.
2. Know the roles and responsibilities for data workloads and common storage formats.

02

TOPIC 02

Relational Data on Azure

Relational design, SQL, Azure SQL and open-source databases

Key Concepts - Relational Data on Azure

1

Relational model

Tables, rows, columns, keys and relationships structure operational data.

2

Normalization

Good design reduces duplication and keeps updates consistent.

3

SQL language

SELECT, JOIN, INSERT, UPDATE and DELETE are the core interaction patterns.

4

Azure choices

Azure SQL, SQL Managed Instance, SQL Server on VMs and open-source services map to different needs.

WHAT YOU WILL DO

Hands-On Labs - Relational Data on Azure

Lab 3

Relational Data, SQL, Normalization, Database Objects
Explain relational data concepts.

You should have a relational diagram, normalization notes, SQL table, object def

Lab 4

Azure SQL and Open-Source Databases on Azure
Describe the Azure SQL family.

You should have comparison tables, scenario answers, managed service benefits, a

Lab 3: Relational Data, SQL, Normalization, Database Objects

LAB 03

Relational Data on Azure

The retail company stores customers, orders, products, and order lines in a relational database. You must explain how the data is structured and queried.

Objectives

Explain relational data concepts. | Identify tables, rows, columns, keys, and relationships. | Understand normalization.

Validation

You should have a relational diagram, normalization notes, SQL table, object definitions, and example queries.

GUIDED LAB FLOW

Lab 3: Step-by-Step

1

Draw a Relational Model

Create entities: Customers

Orders

OrderItems

Products Add primary keys and foreign keys.

2

Explain Normalization

Answer: 1. Why should product data not be repeated on every order line?

2. How do separate tables reduce update errors?

3. What is the tradeoff of normalization?

3

Identify SQL Statement Types

Create a table: | SQL Statement | Purpose |

| --- | --- |

| SELECT | Read data |

| INSERT | Add data |

| UPDATE | Modify data |

| DELETE | Remove data |

| CREATE | Create

4

Review Database Objects

Define: Table

View

Index

Stored procedure

Schema

Primary key

Foreign key

5

Write Example Queries

SELECT ProductName, Price

FROM Products; SELECT c.CustomerName, o.OrderDate

FROM Customers c

JOIN Orders o ON c.CustomerID = o.CustomerID;

CHECKPOINT

Lab 3: Validate and Review

Test it

You should have a relational diagram, normalization notes, SQL table, object definitions, and example queries.

Checkpoint questions

1. What is a primary key?
2. What is a foreign key?
3. Why is normalization used?
4. What does SELECT do?

Exam focus: DP-900 expects recognition of relational concepts and common SQL statements, not advanced SQL programming.

Lab 4: Azure SQL and Open-Source Databases on Azure

LAB 04

Relational Data on Azure

The retail company wants to migrate existing relational databases to Azure. Some apps use SQL Server, while others use PostgreSQL and MySQL.

Objectives

Describe the Azure SQL family. | Compare Azure SQL Database, Azure SQL Managed Instance, and SQL Server on Azure VMs. | Identify open-source database services on Azure.

Validation

You should have comparison tables, scenario answers, managed service benefits, and security/cost notes.

GUIDED LAB FLOW

Lab 4: Step-by-Step

1

Compare Azure SQL Options

Service	Best Fit
Azure SQL Database	Modern cloud app needing managed PaaS database
Azure SQL Managed Instance	SQL Server compatibility with

2

Compare Open-Source Database Services

Document: Azure Database for PostgreSQL

Azure Database for MySQL

Azure Database for MariaDB legacy considerations

3

Match Migration Scenarios

Choose a service: 1. New cloud-native app using SQL Server engine.

2. Existing SQL Server app needing instance-level compatibility.

3. Legacy app requiring OS-level contr

4

Review Managed Service Benefits

Write how managed database services help with: - Patching.

- Backup.
- High availability.
- Scaling.
- Monitoring.
- Security.

5

Security and Cost Notes

Document: Authentication

Firewall or private access

Backup retention

Performance tier

Scaling choice

CHECKPOINT

Lab 4: Validate and Review

Test it

You should have comparison tables, scenario answers, managed service benefits, and security/cost notes.

Checkpoint questions

1. When is Azure SQL Database appropriate?
2. Why choose SQL Server on Azure VMs?
3. What is a managed database service?
4. Which Azure services support PostgreSQL and MySQL?

Exam focus: Know the Azure relational database service families and when each option fits.

TOPIC RECAP

Recap - Relational Data on Azure

1. DP-900 expects recognition of relational concepts and common SQL statements, not advanced SQL programming.
2. Know the Azure relational database service families and when each option fits.

03

TOPIC 03

Non-Relational Data on Azure

Azure Storage, Blob, Files, Tables and Cosmos DB

EXAM FOCUS | Non-relational services

Key Concepts - Non-Relational Data on Azure

1. **Object storage**
   Blob Storage keeps unstructured objects such as images, logs and documents.

2. **Shared files**
   Azure Files provides managed SMB/NFS file shares for lift-and-shift workloads.

3. **Key-value tables**
   Azure Table Storage supports simple NoSQL entity lookups at scale.

4. **Cosmos DB**
   Cosmos DB provides globally distributed NoSQL APIs for document, key-value, graph and column-family patterns.

WHAT YOU WILL DO

Hands-On Labs - Non-Relational Data on Azure

Lab 5

Azure Storage, Blob, Files, Tables

Describe Azure Blob Storage.

You should have storage mappings, concept definitions, and redundancy notes.

Lab 6

Azure Cosmos DB, NoSQL, APIs

Explain non-relational data concepts.

You should have NoSQL notes, Cosmos DB capability notes, API mapping, sample doc

Lab 5: Azure Storage, Blob, Files, Tables

LAB 05

Non-Relational Data on Azure

The retail company needs to store product images, shared documents, application logs, and simple key-value data.

Objectives

Describe Azure Blob Storage. | Describe Azure Files. | Describe Azure Table Storage.

Validation

You should have storage mappings, concept definitions, and redundancy notes.

GUIDED LAB FLOW

Lab 5: Step-by-Step

1

Map Storage Needs

Requirement	Azure Storage Service
Product images	Blob Storage
Shared file share	Azure Files
Simple key-value customer preferences	Tabl

2

Review Blob Storage Concepts

Define: Container
Blob
Block blob
Access tier
Lifecycle management

3

Review Azure Files Concepts

Explain: - SMB and NFS file shares.
- Lift-and-shift file server scenarios.
- Shared access for applications.

4

Review Table Storage Concepts

Explain: - Partition key.
- Row key.
- Entity.
- Schemaless design.

5

Compare Redundancy

Write a short note on locally redundant, zone-redundant, and geo-redundant storage.

CHECKPOINT

Lab 5: Validate and Review

Test it

You should have storage mappings, concept definitions, and redundancy notes.

Checkpoint questions

1. When should you use Blob Storage?
2. When should you use Azure Files?
3. What is Azure Table Storage?
4. Why does redundancy matter?

Exam focus: DP-900 expects recognition of Azure Storage services and their common use cases.

Lab 6: Azure Cosmos DB, NoSQL, APIs

LAB 06

Non-Relational Data on Azure

The retail company wants a globally distributed product catalog and user profile store that can handle flexible schemas and low-latency reads.

Objectives

Explain non-relational data concepts. | Describe Azure Cosmos DB capabilities. | Identify common Cosmos DB APIs.

Validation

You should have NoSQL notes, Cosmos DB capability notes, API mapping, sample document, and partition key explanation.

GUIDED LAB FLOW

Lab 6: Step-by-Step

1

Explain NoSQL Fit

Write why NoSQL can help with: - Flexible schema.
- High scale.
- Global distribution.
- Low-latency access.
- JSON document data.

2

Review Cosmos DB Capabilities

Document: Global distribution
Low latency
Throughput with request units
Partitioning
Consistency levels
Multiple APIs

3

Identify Cosmos DB APIs

Create a table: | API | Common Fit |
| --- | --- |
| NoSQL | JSON document applications |
| MongoDB | MongoDB-compatible apps |
| Cassandra | Wide-column workloads |
| Gr

4

Design a Product Catalog Document

Create example JSON: {
  "id": "P1001",
  "category": "Laptop",
  "name": "Training Laptop",
  "price": 1200,
  "tags": ["business", "portable"]
}

5

Review Partitioning

Explain why a good partition key distributes data and workload evenly.

CHECKPOINT

Lab 6: Validate and Review

Test it

You should have NoSQL notes, Cosmos DB capability notes, API mapping, sample document, and partition key explanation.

Checkpoint questions

1. When is Cosmos DB a good fit?
2. What is a request unit?
3. Why does partitioning matter?
4. Which API supports graph workloads?

Exam focus: Know Cosmos DB capabilities, APIs, and common NoSQL use cases.

TOPIC RECAP

Recap - Non-Relational Data on Azure

1. DP-900 expects recognition of Azure Storage services and their common use cases.
2. Know Cosmos DB capabilities, APIs, and common NoSQL use cases.

04

TOPIC 04

Analytics Workloads on Azure

Ingestion, processing, analytical stores, Fabric, Databricks and real-time analytics

Key Concepts - Analytics Workloads on Azure

1 **Ingestion**
Data enters through batch, stream or event-driven paths.

2 **Processing**
Transformation prepares data for reporting, machine learning or operational use.

3 **Analytical stores**
Warehouses, lakehouses and data lakes organize history for large-scale querying.

4 **Modern platforms**
Microsoft Fabric, Azure Databricks and real-time analytics support different analytics patterns.

WHAT YOU WILL DO

Hands-On Labs - Analytics Workloads on Azure

Lab 7

Analytics, Ingestion, Processing, Analytical Data Stores
Describe analytics workload stages.

You should have pipeline diagram, batch/streaming comparison, analytical store n

Lab 8

Microsoft Fabric, Azure Databricks, Real-Time Analytics
Describe Microsoft Fabric at a high level.

You should have Fabric notes, Databricks notes, real-time service mapping, and a

Lab 7: Analytics, Ingestion, Processing, Analytical Data Stores

LAB 07

Analytics Workloads on Azure

The retail company wants daily sales reports, customer behavior analytics, and near-real-time alerting from web clickstream data.

Objectives

Describe analytics workload stages. | Explain data ingestion and processing. | Compare batch and streaming.

Validation

You should have pipeline diagram, batch/streaming comparison, analytical store notes, and service mapping.

GUIDED LAB FLOW

Lab 7: Step-by-Step

1

Map an Analytics Pipeline

Create a flow: Data sources
Ingestion
Raw storage
Transformation
Analytical store
Semantic model
Visualization

2

Compare Batch and Streaming

Mode	Description	Example
Batch	Processes data in groups on a schedule	Daily sales summary
Streaming	Processes events continuously		

3

Identify Analytical Stores

Document: Data lake
Data warehouse
Lakehouse
Semantic model

4

Choose Azure Services

Need	Possible Service
Data ingestion and orchestration	Data Factory or Fabric Data Factory
Big data processing	Azure Databricks
Lakehous	

5

Review Data Transformation

Explain the difference between raw, cleaned, curated, and aggregated data.

CHECKPOINT

Lab 7: Validate and Review

Test it

You should have pipeline diagram, batch/streaming comparison, analytical store notes, and service mapping.

Checkpoint questions

1. What is data ingestion?
2. What is the difference between batch and streaming?
3. What is a data warehouse?
4. What is a lakehouse?

Exam focus: DP-900 analytics questions test high-level pipeline patterns and service selection.

Lab 8: Microsoft Fabric, Azure Databricks, Real-Time Analytics

LAB 08

Analytics Workloads on Azure

The retail company is comparing analytics platforms for reporting, data engineering, machine learning, and real-time dashboards.

Objectives

Describe Microsoft Fabric at a high level. | Describe Azure Databricks use cases. | Identify real-time analytics services.

Validation

You should have Fabric notes, Databricks notes, real-time service mapping, and architecture diagram.

GUIDED LAB FLOW

Lab 8: Step-by-Step

1

Review Microsoft Fabric Concepts

Document: OneLake
Lakehouse
Warehouse
Data Factory
Real-Time Intelligence
Power BI

2

Review Azure Databricks Concepts

Explain how Databricks can support: - Apache Spark processing.
- Data engineering notebooks.
- Machine learning workloads.
- Delta Lake patterns.
- Collaborative analytic

3

Identify Real-Time Analytics

Match: | Requirement | Possible Service |
| --- | --- |
| Ingest event streams | Event Hubs |
| Process streaming data | Stream Analytics |
| Real-time analytics in Fabri

4

Compare Services

Create a decision table: | Scenario | Better Fit |
| --- | --- |
| Business reporting workspace | Microsoft Fabric |
| Spark-heavy engineering workloads | Azure Databrick

5

Draw an Analytics Architecture

Point of sale -> Event stream -> Real-time processing -> Dashboard
Daily exports -> Lakehouse -> Transformations -> Power BI report

CHECKPOINT

Lab 8: Validate and Review

Test it

You should have Fabric notes, Databricks notes, real-time service mapping, and architecture diagram.

Checkpoint questions

1. What is Microsoft Fabric?
2. What is Azure Databricks commonly used for?
3. What is real-time analytics?
4. Which service can ingest event streams?

Exam focus: Know the high-level purpose of Fabric, Databricks, and Microsoft real-time analytics services.

TOPIC RECAP

Recap - Analytics Workloads on Azure

1. DP-900 analytics questions test high-level pipeline patterns and service selection.
2. Know the high-level purpose of Fabric, Databricks, and Microsoft real-time analytics services.

05

TOPIC 05

Power BI and Exam Readiness

Models, visualizations, service mapping and capstone review

EXAM FOCUS | Visualization and synthesis

Key Concepts - Power BI and Exam Readiness

1

Semantic model

Tables, relationships, measures and hierarchies shape how reports answer questions.

2

Visual design

Charts should match the question, data grain and audience decision.

3

Service matching

DP-900 rewards knowing what each Azure data service is for.

4

Study plan

A confidence matrix and mistakes log focus final revision time.

WHAT YOU WILL DO

Hands-On Labs - Power BI and Exam Readiness

Lab 9

Power BI, Data Models, Visualizations

Describe Power BI capabilities.

You should have Power BI component notes, data model design, visualization choic

Lab 10

DP-900 Capstone and Exam Readiness

Review all DP-900 skill areas.

You should have a service map, architecture diagram, confidence matrix, study pl

Lab 9: Power BI, Data Models, Visualizations

LAB 09

Power BI and Exam Readiness

The retail company wants a dashboard for sales, products, regions, and trends. You must explain how Power BI supports data analysis.

Objectives

Describe Power BI capabilities. | Explain data models and semantic models. | Choose appropriate visualizations.

Validation

You should have Power BI component notes, data model design, visualization choices, and dashboard rules.

GUIDED LAB FLOW

Lab 9: Step-by-Step

1

Identify Power BI Components

Document: Power BI Desktop
Power BI service
Dataset or semantic model
Report
Dashboard
Workspace
Data refresh

2

Design a Simple Data Model

Create tables: Sales
Products
Customers
Dates
Regions Identify relationships and keys.

3

Choose Visualizations

Question	Visualization
Sales by month	Line chart
Sales by product category	Bar chart
Sales by region	Map or filled map
Market shar	

4

Add Measures and Filters

Define examples: Total Sales
Order Count
Average Order Value
Year
Region
Product Category

5

Review Good Dashboard Design

Write five rules for clear dashboards, such as consistent labels, useful filters, and avoiding clutter.

CHECKPOINT

Lab 9: Validate and Review

Test it

You should have Power BI component notes, data model design, visualization choices, and dashboard rules.

Checkpoint questions

1. What is Power BI used for?
2. What is a semantic model?
3. Why are relationships important?
4. Which visualization is suitable for trends over time?

Exam focus: Power BI questions test capabilities, data models, and choosing appropriate visualizations.

Lab 10: DP-900 Capstone and Exam Readiness

LAB 10

Power BI and Exam Readiness

The retail company asks for a data platform recommendation that supports transactions, product images, customer profiles, analytics, real-time events, and dashboards.

Objectives

Review all DP-900 skill areas. | Design an end-to-end Azure data platform. | Match data scenarios to Azure services.

Validation

You should have a service map, architecture diagram, confidence matrix, study plan, and cleanup confirmation.

GUIDED LAB FLOW

Lab 10: Step-by-Step

1

Build a Service Map

Requirement	Data Type or Workload	Azure Service
Online orders	Transactional relational	Azure SQL Database
Product images	Unstructu	

2

Draw the Architecture

Include: Transactional applications

Operational databases

Storage accounts

Ingestion pipelines

Analytical store

Power BI reports

Real-time event path

3

Review DP-900 Skill Areas

Rate confidence:

Skill Area	Confidence 1-5	Study Action
Core data concepts		
Relational data on Azure		
Non-relational data		

4

Build an Exam Mistakes Log

For missed questions, record: Topic:

Wrong assumption:

Correct concept:

Service to review:

5

Clean Up Resources

If you created resources: az group delete --name rg-dp900-lab --yes --no-wait Confirm with your instructor before deleting shared resources.

6

Create a 7-Day Study Plan

1. Day 1: Core data concepts and workload types.
2. Day 2: Relational concepts and SQL.
3. Day 3: Azure SQL and open-source databases.
4. Day 4: Azure Storage and Cosmos

CHECKPOINT

Lab 10: Validate and Review

Test it

You should have a service map, architecture diagram, confidence matrix, study plan, and cleanup confirmation.

Checkpoint questions

1. Which Azure data service is easiest for you to identify?
2. Which service names are most confusing?
3. What is the difference between transactional and analytical data?
4. What will you review before exam day?

Exam focus: DP-900 rewards service recognition and concept clarity. Focus on what each data service is for, not deep administration tasks.

TOPIC RECAP

Recap - Power BI and Exam Readiness

1. Power BI questions test capabilities, data models, and choosing appropriate visualizations.
2. DP-900 rewards service recognition and concept clarity. Focus on what each data service is for, not deep administration tasks.

WRAP-UP

Course Summary and Next Steps

COURSE OUTCOMES

What You Achieved

1

Core Data Concepts

Data types, workloads, roles, stores and file formats

2

Relational Data on Azure

Relational design, SQL, Azure SQL and open-source databases

3

Non-Relational Data on Azure

Azure Storage, Blob, Files, Tables and Cosmos DB

4

Analytics Workloads on Azure

Ingestion, processing, analytical stores, Fabric, Databricks and real-time analytics

5

Power BI and Exam Readiness

Models, visualizations, service mapping and capstone review

NEXT STEPS

DP-900 Exam Preparation

1. Review each lab's checkpoint questions and Exam Focus note.
2. Practice service selection: Azure SQL, Storage, Cosmos DB, Fabric, Databricks and Power BI.
3. Use the Microsoft DP-900 study guide and Microsoft Learn modules.
4. Maintain an exam mistakes log for confusing service names and workload types.
5. Study guide: https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/dp-900

Courseware and Assessment on the LMS

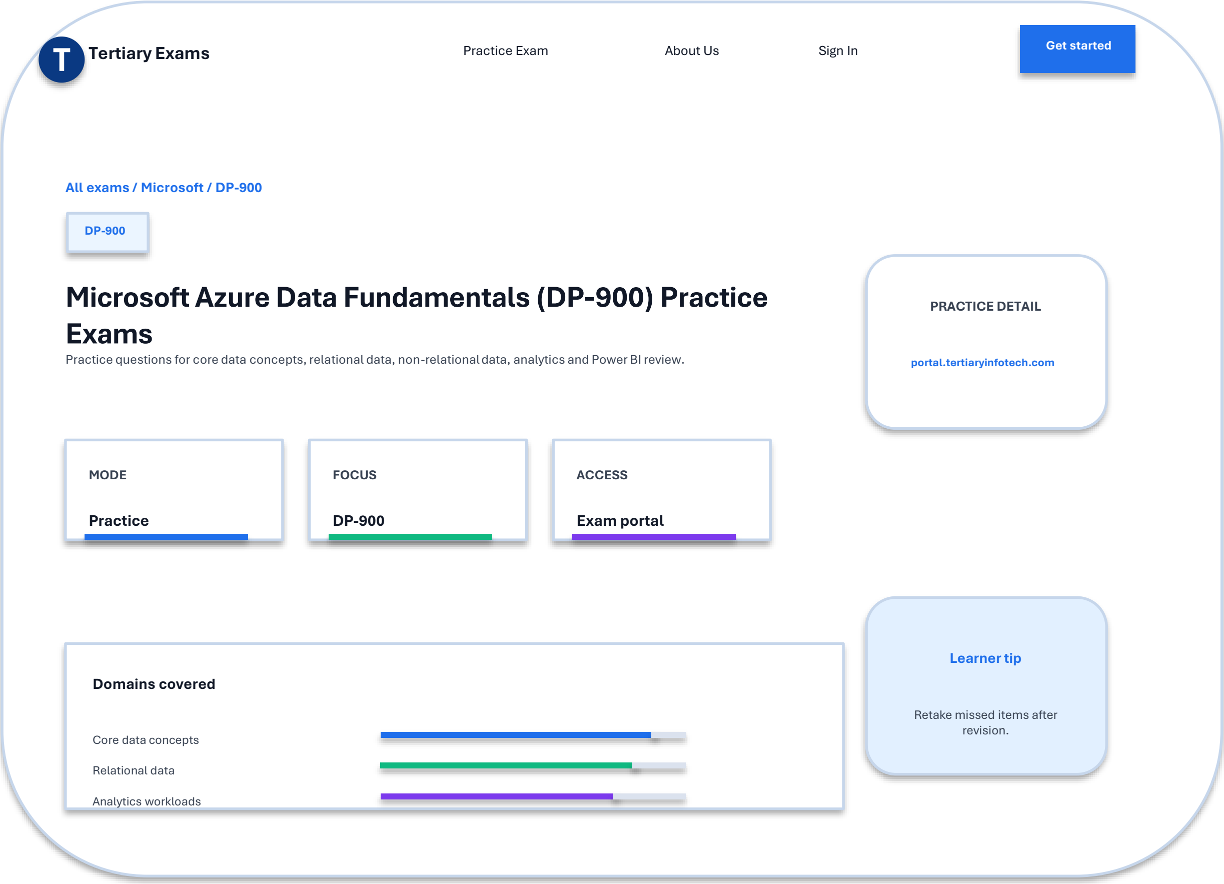
1. Download the courseware from https://lms-tms.tertiaryinfotech.com/
2. Use the Learner Guide during revision and the open-book assessment.
3. Submit assessment answers on the LMS.
4. Complete required feedback and certification steps before leaving.

Take the Online DP-900 Practice Exams

- Reinforce DP-900 topics with realistic, exam-style questions before assessment day.
- Use practice sets to test core data concepts, relational data, non-relational data, analytics workloads and Power BI.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from the Tertiary Exams portal, then bring difficult items to the review session.

Access and practise at:

exams.tertiaryinfotech.com



END REMINDER

Assessment

1. Written Assessment (WA): 1 hour.
2. Practical Performance (PP): 1 hour.
3. Open book: slides, Learner Guide and approved materials only.
4. Take Assessment digital attendance before starting.
5. Submit answers on https://lms-tms.tertiaryinfotech.com/

ASSESSMENT

Assessment Flow Reminder

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then Practical Performance
4. Submit answers on the LMS
5. Sign Assessment Summary Record

The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Digital Attendance (Mandatory)

1. Take assessment digital attendance through the displayed SSG TRAQOM QR code.
2. Confirm the attendance submission before starting assessment.
3. Attendance records support assessment and funding eligibility.
4. Ask the trainer immediately if the QR code or submission fails.

End of Course

You are ready to explain Azure data fundamentals.

Use the labs and Learner Guide to prepare for DP-900.